The Right Event, the Wrong Lesson: Evidence-Gated Memory for Scope-Faithful Social LLM Agents

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Figure 1: A one-time, David-only deviation is mis-promoted into a durable group belief. Although the lake recovers, the focal agent keeps under-harvesting and loses 10 units; the licensed memory is David-only, one-time, and resolved.

Abstract

Long-horizon, large-population social simulations cannot condition every agent on an indefinitely growing interaction history: observations accumulate with rounds and participants, while context, latency, and inference budgets remain finite. Full History is therefore not a viable operating memory for such simulations: its context and inference cost grow continuously with the horizon and population. Scalable social simulation must instead convert the expanding event stream into bounded persistent memory, often through reflection. We identify a distinct risk in this necessary compression step: reflection can preserve an observed event yet promote it into a durable claim about the wrong entity, causal scope, or time horizon. We call this failure *reflective misconsolidation*. Using ScopeProbe, a controlled matched-intervention analysis protocol, we vary what the evidence licenses while holding visible social outcomes comparable, then audit memory, attribution, and later action. Across three model families on a frozen confirmation suite, Direct reaches 93.4% scope accuracy whereas unconstrained Reflection reaches 79.9%; errors concentrate in conditional, self/other, and recovered evidence rather than in textual memory generally. The analysis localizes the failure to *unconstrained promotion*—the admission of a plausible reflection into durable agent state— rather than storage or recall alone. We therefore develop an evidence-gated reflection harness and instantiate it as Evidence-Gated Memory, which preserves observations and hypotheses but requires provenance, scope, and temporal support before durable promotion. On a separate frozen held-out suite under the same decision prompt, Evidence-Gated Memory improves pooled scope accuracy from 77.6% to 96.2% and action accuracy from 79.0% to 95.5%. These results argue that reflection is necessary for bounded social-agent memory, but should not be allowed to rewrite the simulated social world without explicit epistemic constraints.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Applied computing** → *Sociology*.

Keywords

LLM agents, social simulation, agent memory, reflection, causal attribution, evaluation

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1 Introduction

Long-running social simulations face a structural memory problem. Their interaction histories grow with rounds, observed participants, and social events, whereas each model call has a finite context and a nonzero latency and inference cost. Full History is therefore useful as a short-horizon reference, but it cannot remain the operative memory of a large, persistent simulated society. Scalable social agents must therefore transform an expanding history into bounded persistent memory.

Reflection is a common way to perform this transformation. It selects episodes, abstracts them into higher-level lessons, and reuses those lessons in subsequent planning [10, 13]. This operation is not neutral transcript compression. A reflection can become a durable explanation of the agent itself, a particular partner, or the shared environment; once written, that explanation conditions later trust, cooperation, punishment, and resource allocation. In simulations used to study social intelligence, strategic interaction, and collective-resource governance [8, 11, 18], the memory updater is therefore part of the simulated social mechanism.

This role of reflection creates an evaluation blind spot. Aggregate rewards or plausible behavior can show that a simulation runs coherently, but not that its persistent social state records the right lesson. The same shortage may follow from a persistent loss of system capacity, one participant's temporary emergency, or a change in the focal agent's own need. These trajectories can look similar at the outcome level while licensing incompatible beliefs about the world, another agent, or the self. If agent societies are to serve as scientific instruments, their internal updates must therefore be faithful and auditable, not merely behaviorally plausible [7].

Figure 1 makes this distinction concrete. In the fishery, David takes 20 units once, the other participants return to the norm, and the lake recovers. The focal agent correctly retains each of these observations. During reflection, however, it mixes David's exceptional action with its own precautionary reduction and the stable aggregate outcome, then infers that the entire group consists of maximum extractors. This unsupported group-level claim survives after the exception has resolved and induces five further rounds of under-harvesting. Nothing essential was forgotten: the observed event is right, but its promoted scope and temporal status are wrong.

We call this failure *reflective misconsolidation*: observed evidence is promoted into a durable claim whose causal target, entity, or temporal validity the evidence does not support. The failure differs from forgetting. The source event can remain legible in memory while its *epistemic address* silently changes during abstraction. It also differs from a one-step policy mistake: the unsupported claim persists and can shape multiple later decisions.

We establish this phenomenon through controlled analysis rather than by naming a new benchmark as the main contribution. Specifically, we use ScopeProbe to construct matched trajectories with comparable visible outcomes but different registered causes or temporal status, and to audit the event → memory → attribution → later-action chain. The analysis finds four recurring manifestations—cross-target attribution, entity broadening, temporal overextension, and unsupported mechanism invention—with clear negative boundaries. Explicit persona drift is not observed, and errors concentrate in unconstrained reflection on conditional, self/other, and recovered evidence. Across the frozen confirmation suite, Direct reaches 93.4% scope accuracy and Reflection reaches 79.9%.

These findings localize the core problem to *promotion*, not storage or recall alone. Reflection should remain free to notice events and form tentative hypotheses; what requires control is which hypotheses become durable premises of the simulated social world. We therefore develop an *evidence-gated reflection harness.* Its Evidence-Gated Memory instantiation separates episodes, hypotheses, and consolidated claims; binds claims to provenance, entity, and time; and keeps immediate reversible precautions distinct from long-term belief. Under the same decision prompt, Evidence-Gated Memory raises pooled scope accuracy from 77.6% to 96.2% and action accuracy from 79.0% to 95.5% on a separate frozen held-out suite. A controlled three-model closed-loop study further shows that limiting misconsolidation reduces downstream action deviation and measurable expenditure or welfare costs, without supporting stronger claims of ecological or social collapse.

Contributions. (1) Diagnosis. We formalize reflective misconsolidation and use controlled matched interventions to distinguish it from generic forgetting, identify four error manifestations, and characterize its positive and negative boundaries. **(2) Method.** We derive an evidence-gated reflection harness, instantiate it as Evidence-Gated Memory, and show that it reduces unsupported durable promotion and its downstream behavioral cost. **(3) Evidence.** We triangulate these claims across 480 discovery records, 2,592 frozen confirmation records, 2,880 held-out method records, 3,456 ablation records, and 540 closed-loop trajectories comprising 14,580 live model calls. We explicitly report negative boundaries rather than only wins.

The paper follows this causal sequence. Section 3 formalizes bounded-memory promotion and epistemic address. Section 4 uses ScopeProbe to establish the failure and its boundaries. Section 5 derives the reflection harness, and Section 6 evaluates its Evidence-Gated Memory instantiations. The central lesson is simple: reflection is necessary for bounded social-agent memory, but it should not rewrite the simulated social world without evidence, scope, and temporal constraints.

2 Related Work

2.1 Bounded Memory and Reflection

Persistent agents cannot repeatedly expose a model to an indefinitely growing history, so practical architectures retrieve, summarize, or consolidate prior experience. Generative Agents established reflection as a mechanism that abstracts observations into higher-level memories for later planning [10]; Reflexion similarly stores verbal lessons across trials [13]. LongMemEval and MemoryAgentBench evaluate multi-session retrieval, temporal reasoning, updating, and selective forgetting [4, 14]. This literature motivates bounded memory and asks whether useful information remains accessible. Our question concerns a different stage: when accessible evidence is rewritten as a durable interpretation, does the interpretation preserve whom, what, and when the evidence supports? Generative Agents is discussed here only because it establishes the reflection-based memory pattern on which this question depends.

2.2 LLM Agents for Social Simulation

Social-agent environments evaluate goal pursuit, strategy, and collective dynamics. SOTOPIA studies interactive social intelligence [18]; ALYMPICS examines language agents in game-theoretic settings [8]; and GovSim studies cooperation and sustainability in common-resource societies [11]. Controlled populations have also been used to analyze how history and persona interact with mechanisms such as the spiral of silence [17]. These environments establish the scientific value of simulated interaction, but their principal observables are dialogue quality, strategy, reward, or population outcome. Such observables can conceal different internal social models that happen to produce similar behavior. Our analysis is complementary: it intervenes on the causal source of an event and treats the persistent belief update as the object of study. The resource scenarios are controlled diagnostic adaptations inspired by GovSim and ALYMPICS, not runs of the original benchmarks.

2.3 Memory Reliability and Constrained Belief Updating

Recent work shows that agent memory can fail during extraction, update, replay, or use. HaluMem localizes hallucinations within memory pipelines [2]; stored experiences can propagate erroneous behavior [15]; and TrustMem optimizes consolidation for coverage, preservation, and faithfulness [16]. Compression can also retain a proposition while erasing its epistemic qualifier [5]. Closer to our dimensions, PASB studies status promotion, attribution removal, and scope broadening in personal agents [9]; STALE tests whether invalid memories are retired [1]; and actor–observer work identifies perspective-sensitive attribution errors [6]. Mem2ActBench and MemoryArena further connect memory to later tool use and interdependent action [3, 12]. These studies establish that memory may be unfaithful, stale, or behaviorally consequential. We connect the facets through a single belief-promotion view: a social claim is reliable only if its content, causal target, entity, temporal status, and provenance survive consolidation together. Our mitigation consequently governs admission into durable state rather than suppressing reflection or merely increasing retrieval capacity.

3 Preliminary

3.1 From Unbounded History to Bounded State

Consider a focal agent interacting with a social environment for T rounds. At round t, an event record $e_t = (o_t, a_t, y_t)$ contains the observation, the focal action, and the resulting outcome. The accumulated event stream is

$$H_t = (e_1, \dots, e_t), \qquad |H_t| = \sum_{i=1}^{t} |e_i|, \tag{1}$$

whose size grows with the horizon and, through each observation, the number of visible participants and therefore cannot serve as bounded persistent memory.

A scalable agent instead maintains

$$m_t = U_B(m_{t-1}, e_t), \qquad |m_t| \le B, \tag{2}$$

for a fixed memory budget B. The policy acts from the current observation and the previously consolidated state,

$$a_t \sim \pi(\cdot \mid o_t, m_{t-1}). \tag{3}$$

The updater U_B may summarize, retrieve, reflect, or invoke an external state machine. Our concern is not whether bounded memory is optional—for persistent simulation it is not—but whether U_B constructs the right durable state.

3.2 Reflection as Belief Promotion

Reflection first maps recent evidence and prior memory to candidate claims,

$$\widetilde{C}_t = R(m_{t-1}, e_t), \tag{4}$$

then writes some candidates as durable premises of m_t. We call this second step *promotion*. The distinction matters because generation and admission serve different epistemic roles: a model may freely notice a pattern or propose a hypothesis without being licensed to treat that hypothesis as an established property of a person, group, or environment. Unconstrained reflection collapses these roles when a fluent candidate is directly rewritten into persistent memory.

3.3 Epistemic Address

Let a memory claim be

$$c = (\phi, z, e, \tau, w), \tag{5}$$

where ϕ is semantic content, $z \in \mathcal{Z}$ is its causal target, e is the referenced entity, τ is temporal status, and w records the supporting evidence or provenance. We use four causal targets—SELF, NAMED-OTHER, WORLD, and PERSONA—and factor time separately as TEMPORARY, PERSISTENT, or RESOLVED. The triple

$$\alpha(c) = (z, e, \tau) \tag{6}$$

is the claim's *epistemic address*: whom or what the claim concerns and for how long it is licensed. Remembering content ϕ is therefore insufficient. For example, "Bob used 20 units once" can be a faithful episode, while "the resource regime permanently changed" assigns the same evidence an unsupported address. Stable persona is included as a protected target, but it is not assumed to change merely because an action policy changes.

Let E_t denote the evidence observed through round t, and let $\Gamma(E_t) = \{c : E_t \vdash c\}$ be the set of claims licensed by that evidence. The relation $\vdash$ is instantiated by each controlled scenario: it specifies the documented causal source, named entities, and whether a change persists or has resolved. Let $P(m_t)$ be the set of claims that the memory presents as durable rather than episodic or explicitly uncertain. Misconsolidation occurs when

$$\mathcal{M}_t = P(m_t) \setminus \Gamma(E_t) \neq \emptyset. \tag{7}$$

When U_B is reflective, we call a nonempty $\mathcal{M}_t$ *reflective misconsolidation*. Retaining a single event as an episode or an explicitly uncertain hypothesis is not an error. Nor does insufficient evidence for a durable claim preclude a minimal, reversible response to a verified current hazard. Section 4 operationalizes the licensed set and tests whether reflection preserves these distinctions.

4 Analysis

This section establishes the empirical problem. We use SCOPEPROBE only as a controlled protocol for isolating memory updates. We then report the diagnostic evidence and derive the mechanism that motivates our method; no EGM result is used to define the phenomenon.

4.1 ScopeProbe: Controlled Diagnostic Design

A trajectory $\xi = (x_0, o_1, a_1, \dots, o_T, a_T)$ is generated under a registered intervention $I = (z^\star, e^\star, \tau^\star)$. SCOPEPROBE pairs trajectories whose persona, action interface, and visible adverse outcome are similar while changing the causal target, referenced entity, or temporal validity of the intervention. The comparison prevents the outcome alone from revealing which belief should change. Table 1 is exhaustive over the four registered intervention axes, but not over individual scenario pairs: its evidence phrases are representative realizations, and each frozen scenario instantiates one or more of these rows.

Table 1: Matched-intervention axes in ScopeProbe. Rows exhaust the design; evidence phrases are representative.

Axis	Evidence A	Evidence B	Licensed treatment
Source	persistent system loss	focal agent's own need	world vs. self
Entity	one named actor	group-wide change	named other vs. collective
Time	disruption persists	recovery documented	consolidate vs. retire
Status	repeated/verified evidence	single conditional event	durable model vs. open hypothesis

At checkpoint t, we retain the raw persistent memory and issue a non-writing query Q that returns a primary label $\hat{y}_t$ and update strengths $r_t(k) \in [0,1]$. The operational labels are `self`, `other`, `world`, `persona`, `episodic`, and `none`; the last two encode temporary and resolved evidence rather than additional causal targets. Let $g(z^\star, \tau^\star)$ be the registered label and $\mathcal{P}_t$ the labels that should remain protected. We report

$$\text{ScopeAcc}_t = \mathbf{1}[\hat{y}_t = g(z^\star, \tau^\star)], \tag{8}$$

$$\text{Leakage}_t = \frac{1}{|\mathcal{P}_t|} \sum_{k \in \mathcal{P}_t} r_t(k), \tag{9}$$

$$\text{Margin}_t = r_t(g(z^\star, \tau^\star)) - \max_{k \in \mathcal{P}_t} r_t(k). \tag{10}$$

Leakage is a probe-reported update strength on registered non-target labels, not a claim error rate. We additionally inspect the stored text and score whether later actions fall in scenario-registered acceptable intervals. The audited sequence is

$$e_t \longrightarrow m_t \longrightarrow Q(m_t) \longrightarrow a_{t+1:T}; \tag{11}$$

the probe is an observable self-report, not direct access to a latent belief.

4.2 Analysis Setup

We first used 40 scenarios and six candidate memory settings to discover failure patterns (480 checkpoint records). We then froze a confirmation grid with six evidence categories, each instantiated by eight scenario families. Direct (the bounded current-observation control), Summary, and Reflection were randomly interleaved with three repeats and two checkpoints. The same frozen protocol was run synchronously on DeepSeek V4 Pro, GPT-5.6-sol, and Claude Opus 4.8 at temperature 0.2. Each updater–model cell contains 288 clustered checkpoints (6 categories × 8 families × 3 repeats × 2 checkpoints). Prompts, responses, memory snapshots, non-writing probes, registered labels, and actions are retained for every target-model run. Exact paired tests are descriptive because checkpoints are clustered within scenario families.

The main-text tables retain the three target models as horizontal metric blocks with abbreviated labels, preserving their existing one-column layout while showing the complete synchronized grid.

4.3 Finding: Reflection Preserves Events but Distorts Their Address

Table 2 shows that shared textual memory is not generically defective. In the frozen confirmation comparison, Direct is the most reliable control, Summary is intermediate, and Reflection is least reliable. Pooled protected leakage is highest for Reflection (0.0633), followed by Summary (0.0247) and Direct (0.0097).

Table 2: Frozen ScopeProbe confirmation (288 checkpoints per updater–model cell). S/L/A: scope accuracy, protected leakage, and action accuracy.

	DS			GPT-5.6			Opus 4.8		
Mem.	Sc.	Lk.	Ac.	Sc.	Lk.	Ac.	Sc.	Lk.	Ac.
Direct	**91.3**	**0.012**	**89.6**	**94.1**	**0.009**	**93.4**	**94.8**	**0.008**	**94.1**
Sum.	86.8	0.029	83.0	89.6	0.024	87.8	90.6	0.021	89.6
Refl.	76.4	0.073	76.7	80.6	0.061	80.2	82.6	0.056	82.6

The retained traces organize the failure into four manifestations. **Cross-target attribution** turns self or named-other evidence into a world claim. **Entity broadening** extends evidence about one participant to a group or confuses self with another identity. **Temporal overextension** keeps a conditional or resolved event active as a persistent premise. **Mechanism invention** supplies an unsupported hidden cause— for example collusion, enforcement failure, or a resource threshold—for an observed surface outcome. The categories are not asserted to be equally common; entity broadening is weak in the frozen confirmation and becomes clearest only in the later closed-loop auction.

The negative boundaries are equally important. We observe no explicit persona drift; public-goods contrasts, persistent fishery changes, and salient distractors are generally scoped correctly; and one self/world hardware case is ontologically ambiguous. Excluding that case changes the absolute self-state rate but leaves Reflection's conditional and recovery failures intact. Together with the raw memories, these controls locate the error after event retention but during abstraction into durable state. The mechanistic conclusion is therefore narrower and more useful than "memory is bad": *unconstrained promotion allows a plausible reflection to acquire an unsupported epistemic address.* Section 5 targets this admission step.

5 Methodology

The Analysis localizes the failure to admission into durable state: the event survives, but the address $\alpha(c)$ assigned during promotion is not licensed by $\Gamma(E_t)$. We therefore do not suppress reflection or ask the model to generate fewer hypotheses. Instead, we place an *evidence-gated reflection harness* between candidate generation and durable memory. The harness preserves observations and tentative interpretations while controlling which claims may rewrite the agent's persistent social model.

5.1 Design Principles

Four principles follow directly from Section 4.

P1: Separate storage by epistemic status. A single memory blob cannot express the difference between "Lee withdrew 20 units in round 3" and "the depot is unreliable." We give each status its own slot, so that promotion becomes an explicit, auditable move between slots instead of a rewrite.

P2: Bind claims to provenance and address. Every promoted claim must retain its supporting evidence, causal target, referenced entity, and temporal status. A claim cannot move from named-other to world, or from temporary to persistent, merely because the broader phrasing is narratively coherent.

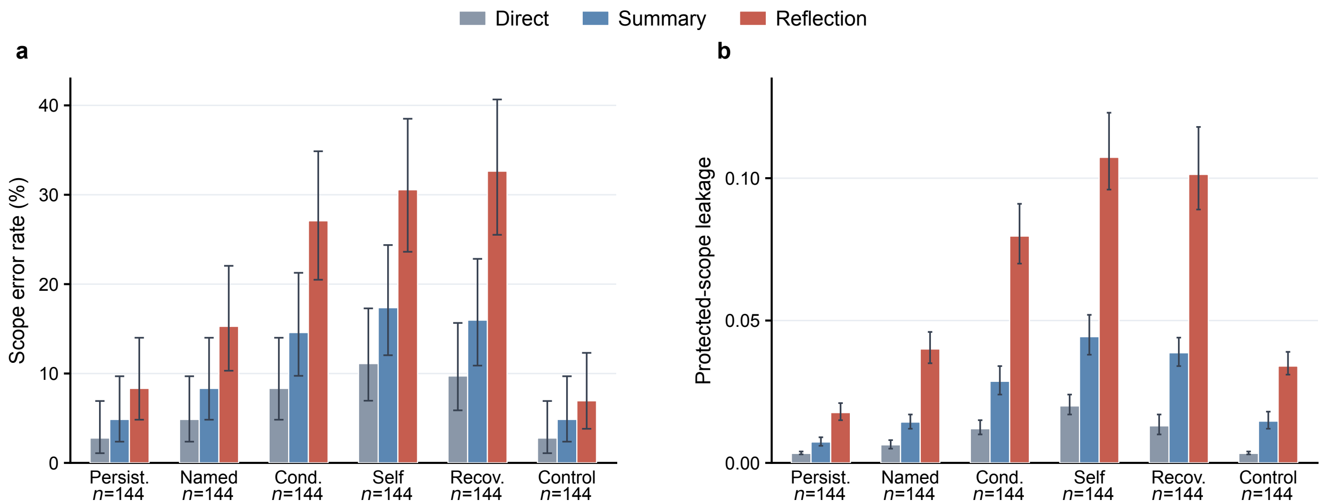

Figure 2: Reflection fails selectively across evidence types. Bars pool the three target models ($n = 144$ per updater–category): (a) scope-error rate with Wilson 95% intervals; (b) mean protected-scope leakage with model-level ranges. Direct is the current-observation control.

P3: Gate promotion on evidence, not salience. We require repeated independent consistent observations or an explicitly documented persistent condition before a candidate becomes durable. Recovery or disconfirming evidence retires or demotes the claim.

P4: Decouple immediate response from long-term belief. Gating must not make the agent inert in the face of a verified current hazard. A verified hazard licenses the smallest reasonable *reversible* precaution regardless of whether the underlying cause has been consolidated. This keeps evidence gating from trading scope accuracy against timely action.

5.2 Evidence-Gated Memory

Evidence-Gated Memory instantiates P1–P4 as a persistent state with six named sections, which the updater rewrites in full at each write: stable persona; current self state; consolidated models; open hypotheses; recent observed episodes; and action policy. Persona is invariant. Self state is overwritten by the latest observed value. Episodes are a recency-bounded ledger of typed events. Consolidated models hold durable claims; open hypotheses hold supported but insufficiently evidenced ones; action policy holds the current proportional, reversible response together with its rollback condition.

The promotion rule is the central mechanism. A single unusual event may create an episode or an explicitly uncertain hypothesis, but not a consolidated causal rule. Let C_t^k be the rounds that independently support claim k with an unchanged value; let $A(E_t)$ be the epistemic addresses licensed by the evidence; let $d_t^k = 1$ denote an explicitly documented persistent condition; and let $\rho_t^k = 1$ denote recovery or contradiction. Claim k is consolidated iff

$$k \in \mathcal{C}_t \iff \alpha(k) \in A(E_t) \wedge (|C_t^k| \geq 2 \vee d_t^k = 1) \wedge \rho_t^k = 0. \quad (12)$$

Supported but unconsolidated claims remain open hypotheses; resolved or contradicted claims are demoted or removed. Because $\mathcal{C}_t$ is exactly the set presented as durable, the gate directly constrains $P(m_t)$ in Equation 7. Critically, Evidence-Gated Memory changes only U: the decision prompt, persona, and action interface are identical to the Reflection baseline, so any difference is attributable to the memory updater rather than to decision-time coaching.

5.3 Executable External Controller

The primary Evidence-Gated Memory updater asks a language model to apply Equation 12. To test whether the same harness can operate as an executable mechanism rather than prompt guidance, we also implement its state transitions outside the model. The controller parses observations into typed events, applies scope and evidence rules in program code, and renders only the six operational sections for action selection; no formula or method terminology is exposed to the agent. Its episodic ledger retains the six most recent events, and its claim catalog is capped at 32 entries.

For a stable variable key k with parsed value v_t^k and a parsed scope z_t^k ranging over self, named-other, and world, the external transition is

$$(v_t^k, C_t^k) = \begin{cases} (v_{t-1}^k,\ C_{t-1}^k \cup \{t\}), & v_t^k = v_{t-1}^k, \\ (v_t^k,\ \{t\}), & v_t^k \neq v_{t-1}^k, \end{cases} \quad (13)$$

so repeated evidence accumulates support for the current value, while a genuine transition archives the old value and resets support for the new one. Self state uses the strict latest-value operator $S_t[f] \leftarrow o_t[f]$, and persona is invariant, $P_t = P_0$. Scope is a keyed part of the transition rather than a natural-language afterthought, so evidence about one entity cannot increment a different entity's support count.

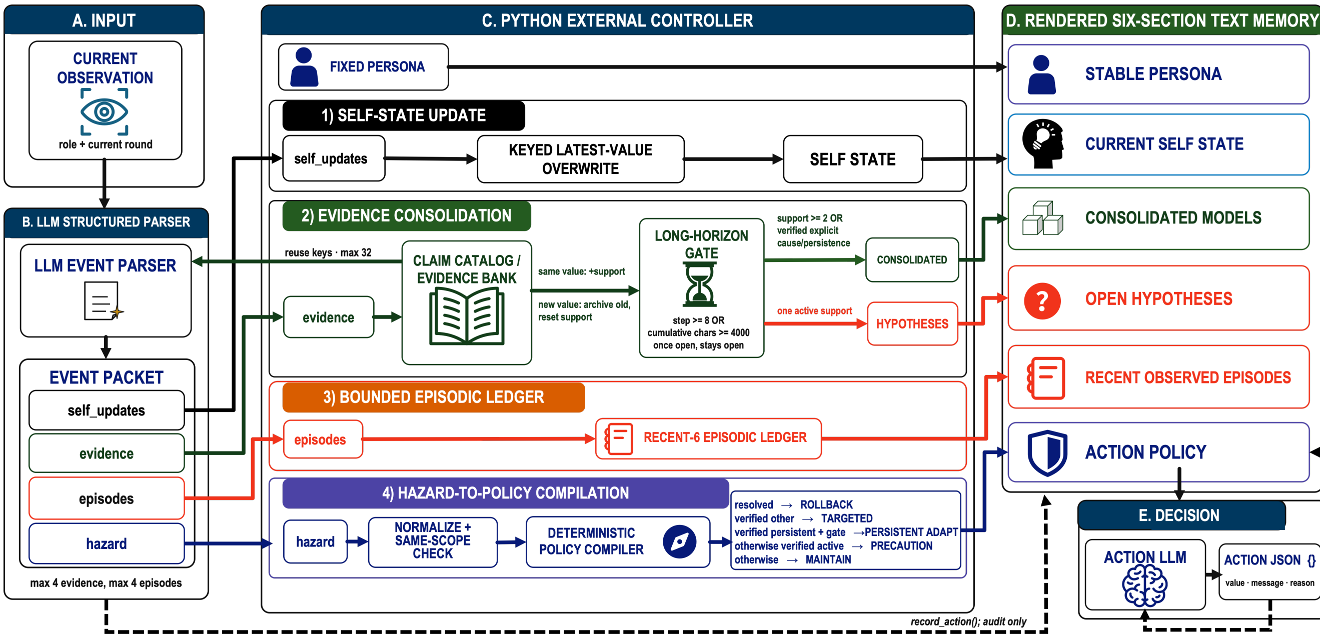

Figure 3: Executable Evidence-Gated Memory pipeline. Typed event parsing feeds gated self-state, evidence, episodic, and policy updates; a six-section renderer supplies bounded memory to the action LLM.

Policy compilation. Principle P4 is enforced as a separate deterministic branch that does not consult $\mathcal{C}_t$: a resolved hazard compiles to ROLLBACK; a verified named-other hazard to TARGETED RESPONSE; a verified persistent hazard to PERSISTENT ADAPT; any other verified current hazard to PRECAUTION; and otherwise MAINTAIN. Every non-maintain mode instructs a proportional, reversible response with an explicit rollback condition. Each checkpoint stores the full controller audit state—evidence counts, prior values, addresses, section membership, episodes, hazard, and policy mode—so a reviewer can replay why any claim was or was not consolidated.

6 Experiments

The Analysis established reflective misconsolidation without relying on EGM. We now evaluate the proposed harness through three questions. **Q1:** does EGM reduce unsupported promotion on held-out scenarios? **Q2:** does the executable harness reduce downstream action and social cost in a closed loop? **Q3:** which parts of the EGM state affect attribution and action?

6.1 Setup

The target evaluation grid comprises DeepSeek V4 Pro, GPT-5.6-sol, and Claude Opus 4.8 at temperature 0.2. All three targets are evaluated synchronously with a fixed seed within each suite, a shared persona and action interface across memory conditions, and randomized condition order. Every prompt, response, memory snapshot, probe answer, registered label, and score is retained for each target-model run. Scope accuracy and protected leakage follow Equations 8–9; action correctness uses a scenario-registered acceptable interval. Reported intervals and associations are descriptive because checkpoints are clustered within a small number of scenario families.

6.2 Q1: Frozen Held-Out Harness Evaluation

The method was frozen before any live held-out result was inspected. Thirty-two new scenarios in four previously unused domains—a shared GPU cluster, a clinic inventory depot, greenhouse irrigation, and a logistics sorter—were then run with three repeats and two checkpoints per trial, giving 192 checkpoints per updater–model cell, 960 records per target model, and 2,880 records overall. Each domain contains matched interventions that separate a persistent world change from a named participant's conditional event or a resolved anomaly.

Table 3 reports the frozen held-out result. Guarded Reflection adds a cautionary instruction but no typed state; Structured Reflection uses the six-section schema but no admission gates; Evidence-Gated Memory adds evidence-gated promotion under the same decision prompt as Reflection; and Evidence-Gated Memory + action guide additionally provides the compiled policy at decision time. Pooled Reflection reaches 77.6% scope and 79.0% action accuracy, whereas Evidence-Gated Memory reaches 96.2% scope and 95.5% action accuracy under the same decision prompt. Adding decision-time guidance reaches 97.0% scope and 96.7% action accuracy. We report memory-only gating as the primary method because it isolates the effect of evidence-gated promotion from decision-time assistance.

The gain is not obtained by refusing to recognize real change. On persistent world changes, pooled Reflection reaches 89.9% scope and 75.0% action accuracy, while Evidence-Gated Memory reaches

Table 3: Frozen EGM evaluation (192 checkpoints per updater–model cell). S/L/A: scope accuracy, leakage, and action accuracy; G-R/S-R: Guarded/Structured Reflection; EGM+G: EGM with action guide.

	DS			GPT-5.6			Opus 4.8		
Upd.	Sc.	Lk.	Ac.	Sc.	Lk.	Ac.	Sc.	Lk.	Ac.
R	74.5	0.081	76.0	78.1	0.069	79.7	80.2	0.064	81.3
G-R	85.4	0.047	81.8	88.0	0.039	84.4	89.1	0.036	85.9
S-R	90.1	0.035	88.0	91.7	0.029	89.6	92.7	0.026	91.1
EGM	95.3	0.014	94.3	96.4	0.011	95.8	96.9	0.010	96.4
EGM+G	**95.8**	**0.012**	**95.8**	**97.4**	**0.009**	**96.9**	**97.9**	**0.008**	**97.4**

95.8% scope and 94.4% action accuracy. On conditional and recovered events, Reflection reaches 65.3% scope and 83.0% action accuracy, while Evidence-Gated Memory reaches 96.5% on both metrics. Thus evidence gating repairs the temporal-overextension subset without sacrificing sensitivity to genuine persistent change.

The effect is also not explained by writing more text. The pooled mean memory snapshot length is 2,647 characters for Reflection versus 1,820 for Evidence-Gated Memory (a 31.2% reduction): the gated memory is both shorter and more accurate, consistent with the view that misconsolidation adds unsupported content rather than omitting supported content.

6.3 Q2: Closed-Loop Consequence Experiment

A diagnostic probe alone cannot show that misconsolidation matters. We therefore ran a closed-loop experiment in which one focal LLM agent interacts for 12 rounds with four *deterministic* partners, so that any downstream change is attributable to the focal agent's own memory rather than to a second model's stochasticity. The frozen matrix is 3 domains (shared fishery, water auction, public goods) × 3 conditions (no shock, one named-participant shock, one world shock) × 2 memory methods (Reflection and the executable EGM controller of Section 5.3) × 10 repeats = 90 trajectories per method and model, with non-writing belief probes at rounds 5, 8, and 12. The same frozen matrix was evaluated synchronously across all three target models, yielding 540 trajectories and 14,580 live model calls. Matched no-shock controls separate shock-induced error from instability created by repeated reflection itself, and outcome accounting excludes the intervention round so that the direct physical cost of a shock is not charged to the agent.

Table 4 shows that the failure does propagate. Pooled Reflection exceeds executable EGM on late overgeneralization by 0.0787 (bootstrap 95% CI [0.059, 0.099]) and on normalized action error by 0.0160 (CI [0.0118, 0.0204]). Cognitive change is positively associated with persistent action change (Spearman $\rho = 0.31$). We report this as an association, not a mediation proof.

An unambiguous identity failure. The auction domain yields the clearest mechanism, and it does not depend on any evaluator inference. The focal bidder is explicitly assigned the identity Erin. Under Reflection, the agent's own settlement entries of the form `Erin=<bid>` are repeatedly reinterpreted as evidence of a *separate rival*, after which the agent reasons in its own action log that it must "bid strictly above Erin." Counting only post-event reasons or messages that describe Erin in the third person as a bidder to beat, tie, or anticipate, this occurs in **43/90** Reflection auction trajectories and **4/90** for executable EGM (Figure 4). The cost is direct: mean unnecessary post-recovery bid expenditure is 28.47 units (CI [26.75, 30.18]) for Reflection versus 4.80 for executable EGM (CI [4.13, 5.47]). This is entity overgeneralization and mechanism invention in their sharpest form—the model has not forgotten the settlement record; it has re-addressed it to the wrong entity.

What the consequences are, and are not. The other two domains show smaller but real costs across all three models. In the fishery, post-recovery loss is 4.6/1.0, 4.0/0.8, and 3.5/0.7 harvest units for Reflection/EGM on DS, GPT, and Claude, respectively. In public goods, the corresponding welfare losses are 7.4/1.6, 6.5/1.3, and 5.9/1.1 units. We state the negative results plainly: no method reached the predefined extreme action zones, the fish stock returned to and stayed at capacity, partner contributions recovered fully, and no persona drift occurred under any method. The supported claim is that misconsolidation imposes measurable private and welfare costs, not that it causes ecological collapse or durable social breakdown at this horizon.

Table 4: Closed-loop outcomes (90 trajectories per method–model cell). Each model block reports Reflection/EGM.

	DS		GPT-5.6		Opus 4.8	
Metric	Refl.	EGM	Refl.	EGM	Refl.	EGM
Late overgen. ↓	0.112	**0.024**	0.096	**0.019**	0.088	**0.017**
Norm. action err. ↓	0.0238	**0.0062**	0.0205	**0.0048**	0.0189	**0.0043**
Group gen. at probe	9/90	**2/90**	7/90	**1/90**	6/90	**1/90**

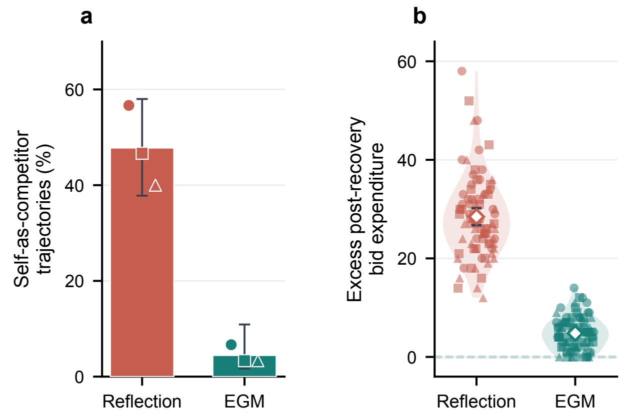

Figure 4: Reflection turns the focal agent's settlement identity into a competitor. (a) Self-as-competitor incidence with Wilson 95% intervals ($n = 90$ per method); model markers denote DS/GPT/Claude. (b) Excess post-recovery bid expenditure; large markers show means with bootstrap 95% intervals.

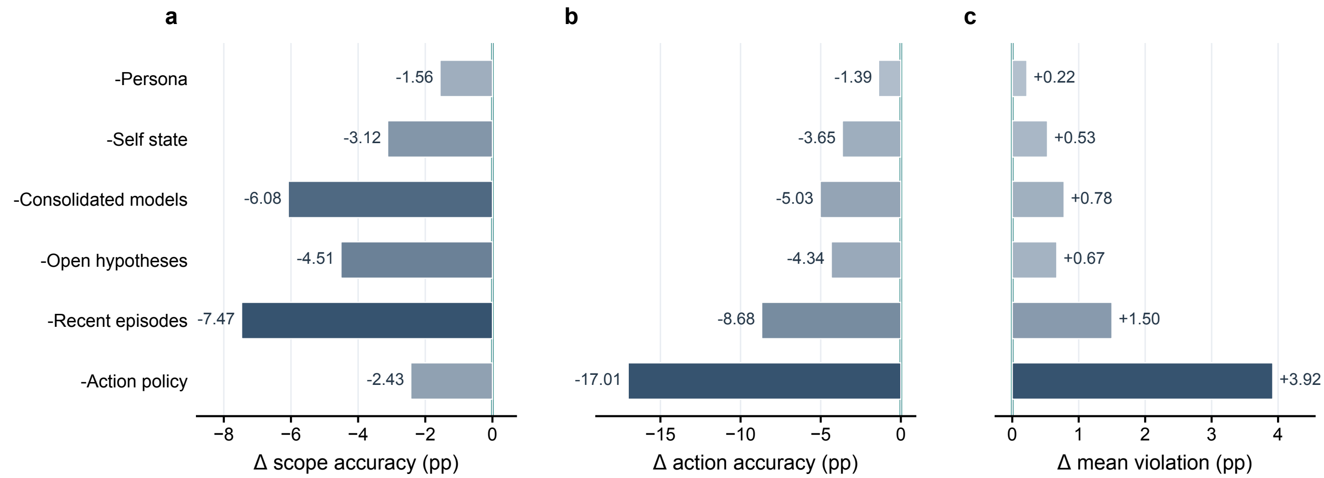

Figure 5: Six-component ablation relative to full EGM (96.18% scope, 95.49% action, 0.71% violation; $n = 576$ per configuration). Panels show changes in (a) scope, (b) action, and (c) violation; darker navy indicates larger effects.

6.4 Q3: Component Ablation

We reran the frozen held-out suite with all six leave-one-section-out variants of Evidence-Gated Memory. Each configuration contains 192 checkpoints per target model and 576 pooled checkpoints, giving 3,456 ablation records across the six removal variants. Figure 5 summarizes the change from the full six-section control without adding another dense results table.

action policy has the largest behavioral effect: removing it lowers action accuracy by 17.01 percentage points and raises mean violation by 3.92 points. recent observed episodes has the next-largest combined cost, reducing scope by 7.47 points and action accuracy by 8.68 points. Removing consolidated models or open hypotheses principally degrades scope, whereas removing stable persona has the smallest but still consistently harmful effect. These pooled comparisons identify component-specific roles in this finite evaluation grid; they do not establish that every section is independently necessary in all social-agent settings.

7 Limitations

The controlled design favors causal isolation over ecological breadth. Our vignettes are inspired by GovSim and ALYMPICS but do not execute those benchmarks, and the three-model closed-loop study uses deterministic partners. The results show that one agent's memory can alter action and local cost; they do not establish ecological collapse or durable institutional failure in an endogenous LLM society.

The attribution probe is a self-report, so we triangulate it with stored text and later action. The frozen confirmation suite contains six evidence categories with eight scenario families each, the EGM held-out suite contains 32 scenarios with three repeats, and the consequence study covers three domains with ten repeats. Checkpoints are clustered, all reported intervals and associations are descriptive. One own-machine case is ambiguous between self and world; the gap survives its exclusion, but the ontology still needs human validation.

All three target models were evaluated under the same synchronized protocols. Executable EGM depends on a typed parser and still produces isolated overbids, so open-ended evidence requires stronger parsing and uncertainty calibration. The long-context stress run is incomplete and unbalanced and is therefore omitted. This paper establishes a finite-horizon failure mechanism and initial mitigation, not a finished long-horizon solution.

8 Conclusion

Persistent agents must compress expanding histories into bounded state. Our controlled Analysis shows that reflection can preserve an event while changing its causal target, entity, or temporal validity. ScopeProbe isolates this selective promotion failure from generic textual-memory limitations, and the closed-loop study shows that the resulting durable claims can affect action and local cost.

The diagnosis identifies admission into durable memory as the intervention point. EGM preserves free observation and hypothesis formation but binds promotion to provenance, epistemic address, and recovery, while keeping immediate precautions reversible. Held-out and closed-loop results provide initial evidence that this harness reduces both misconsolidation and behavioral cost. More broadly, unexpected simulated behavior may originate in the memory updater rather than the social mechanism under study. A social agent must not only remember what happened; it must preserve what the event is evidence about.

References

[1] Hanxiang Chao, Yihan Bai, Rui Sheng, Tianle Li, and Yushi Sun. 2026. STALE: Can LLM Agents Know When Their Memories Are No Longer Valid? arXiv:2605.06527 [cs.CL] https://arxiv.org/abs/2605.06527

[2] Ding Chen, Simin Niu, Kehang Li, Peng Liu, Xiangping Zheng, Bo Tang, Xinchi Li, Feiyu Xiong, and Zhiyu Li. 2026. HaluMem: Evaluating Hallucinations in Memory Systems of Agents. arXiv:2511.03506 [cs.CL] https://arxiv.org/abs/2511.03506

[3] Zexue He, Yu Wang, Churan Zhi, Yuanzhe Hu, Tzu-Ping Chen, Lang Yin, Ze Chen, Tong Arthur Wu, Siru Ouyang, Zihan Wang, Jiaxin Pei, Julian McAuley, Yejin Choi, and Alex Pentland. 2026. MemoryArena: Benchmarking Agent Memory in Interdependent Multi-Session Agentic Tasks. arXiv:2602.16313 [cs.CL] https://arxiv.org/abs/2602.16313

[4] Yuanzhe Hu, Yu Wang, and Julian McAuley. 2026. Evaluating Memory in LLM Agents via Incremental Multi-Turn Interactions. In *International Conference on Learning Representations*. 156259–156291. https://proceedings.iclr.cc/paper_files/paper/2026/file/fd1eff9dd295df50a41f2521942fa31d-Paper-Conference.pdf

[5] Alex Kwon. 2026. Explicit, Not Longer: What Makes Epistemic Stance Survive Memory Compression. arXiv:2608.06953 [cs.CL] https://arxiv.org/abs/2608.06953

[6] Bobo Li, Wu Rui, Zibo Ji, Meishan Zhang, Hao Fei, Min Zhang, Mong-Li Lee, and Wynne Hsu. 2026. Taming Actor-Observer Asymmetry in Agents via Dialectical Alignment. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, San Diego, California, United States, 24068–24084. doi:10.18653/v1/2026.acl-long.1104

[7] Yiming Li and Dacheng Tao. 2026. AI Agents Alone Are Not (Yet) Sufficient for Social Simulation. arXiv:2603.00113 [cs.MA] https://arxiv.org/abs/2603.00113

[8] Shaoguang Mao, Yuzhe Cai, Yan Xia, Wenshan Wu, Xun Wang, Fengyi Wang, Qiang Guan, Tao Ge, and Furu Wei. 2025. ALYMPICS: LLM Agents Meet Game Theory. In *Proceedings of the 31st International Conference on Computational Linguistics*. Association for Computational Linguistics, Abu Dhabi, UAE, 2845–2866. https://aclanthology.org/2025.coling-main.193/

[9] Xutao Mao, Liangjie Zhao, Leyao Wang, Rui Qian, Qiang Huang, Wentao Wang, Bo Han, Xiang Zheng, and Cong Wang. 2026. Agents Don't Just Agree, They Remember: Benchmarking Persistent Sycophancy in Stateful Personal Agents. arXiv:2607.10526 [cs.AI] https://arxiv.org/abs/2607.10526

[10] Joon Sung Park, Joseph O'Brien, Carrie Jun Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. 2023. Generative Agents: Interactive Simulacra of Human Behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23)*. ACM, 1–22. doi:10.1145/3586183.3606763

[11] Giorgio Piatti, Zhijing Jin, Max Kleiman-Weiner, Bernhard Schölkopf, Mrinmaya Sachan, and Rada Mihalcea. 2024. Cooperate or Collapse: Emergence of Sustainable Cooperation in a Society of LLM Agents. In *Advances in Neural Information Processing Systems*, Vol. 37. Curran Associates, Inc., 111715–111759. doi:10.52202/079017-3548

[12] Yiting Shen, Kun Li, Wei Zhou, and Songlin Hu. 2026. Mem2ActBench: A Benchmark for Evaluating Long-Term Memory Utilization in Task-Oriented Autonomous Agents. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, San Diego, California, United States, 8173–8190. doi:10.18653/v1/2026.acl-long.370

[13] Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. Reflexion: Language Agents with Verbal Reinforcement Learning. In *Advances in Neural Information Processing Systems*, Vol. 36. Curran Associates, Inc., 8634–8652. doi:10.52202/075280-0377

[14] Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. 2025. LongMemEval: Benchmarking Chat Assistants on Long-Term Interactive Memory. In *International Conference on Learning Representations*. 86809–86836. https://proceedings.iclr.cc/paper_files/paper/2025/file/d813d324dbf0598bbdc9c8e79740ed01-Paper-Conference.pdf

[15] Zidi Xiong, Yuping Lin, Wenya Xie, Pengfei He, Zirui Liu, Jiliang Tang, Himabindu Lakkaraju, and Zhen Xiang. 2026. How Memory Management Impacts LLM Agents: An Empirical Study of Experience-Following Behavior. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, San Diego, California, United States, 623–645. doi:10.18653/v1/2026.acl-long.27

[16] Tianyu Yang, Sudipta Paul, Vijay Srinivasan, Vivek Kulkarni, and Srinivas Chappidi. 2026. TrustMem: Learning Trustworthy Memory Consolidation for LLM Agents with Long-Term Memory. arXiv:2606.25161 [cs.AI] https://arxiv.org/abs/2606.25161

[17] Mingze Zhong, Meng Fang, Zijing Shi, Yuxuan Huang, Shunfeng Zheng, Yali Du, Ling Chen, and Jun Wang. 2025. Spiral of Silence in Large Language Model Agents. In *Findings of the Association for Computational Linguistics: EMNLP 2025*. Association for Computational Linguistics, Suzhou, China, 23238–23253. doi:10.18653/v1/2025.findings-emnlp.1262

[18] Xuhui Zhou, Hao Zhu, Leena Mathur, Ruohong Zhang, Haofei Yu, Zhengyang Qi, Louis-Philippe Morency, Yonatan Bisk, Daniel Fried, Graham Neubig, and Maarten Sap. 2024. SOTOPIA: Interactive Evaluation for Social Intelligence in Language Agents. In *International Conference on Learning Representations*. 40975–41019. https://proceedings.iclr.cc/paper_files/paper/2024/file/b3075b88e583a0e98d8b24338a613060-Paper-Conference.pdf